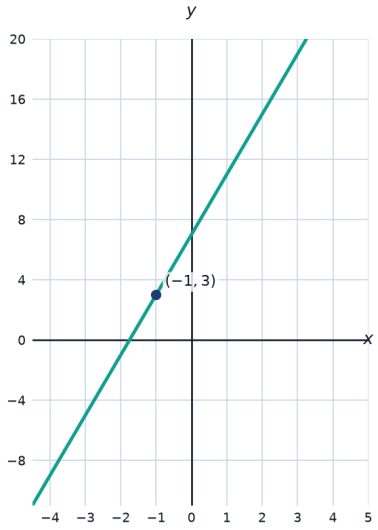
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

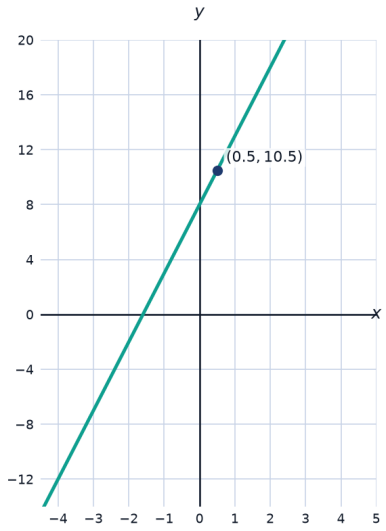
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in negative indices, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 8$ in variables and indices, evaluate $f(2)$.



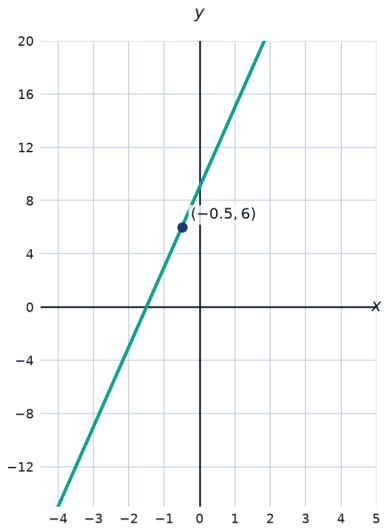
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for variables and indices consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 9$ in fractional indices (extended), evaluate $f(2)$.

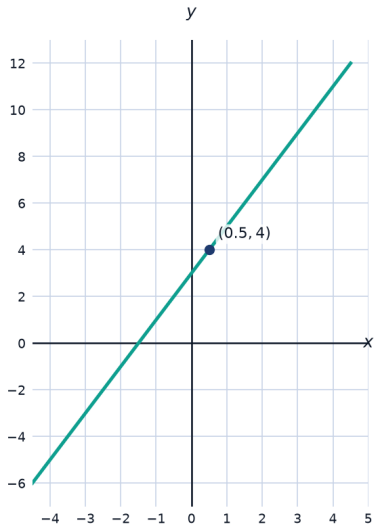


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for fractional indices (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 3$ in standard form, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for standard form consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

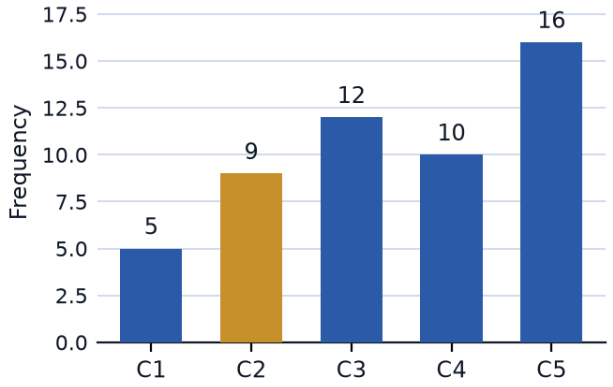
[Statistics & Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A distribution of test scores has first quartile $Q_1 = 58$ and third quartile $Q_3 = 82$. Using the standard $1.5 \times \text{IQR}$ rule, what is the lowest score that would NOT be considered an outlier?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A fair coin is tossed 10 times. Find the probability of getting (a) exactly 6 heads, (b) fewer than 3 heads.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Statistics & Data Analysis · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X < 65) = 0.8413$ and $P(X < 45) = 0.0228$. Find the values of μ and σ.

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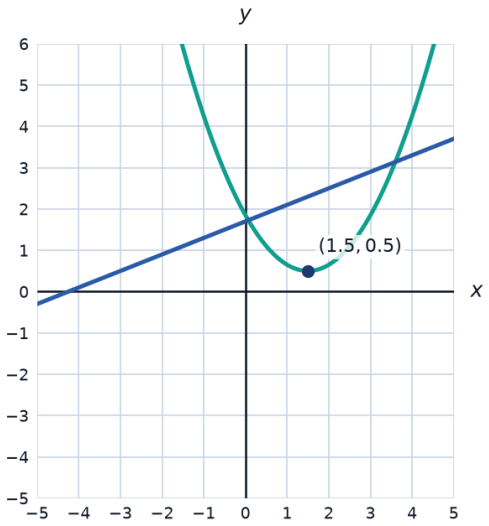
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Mensuration

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p+q\sqrt{5}$, where p and q are rational numbers to be found.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Geometry

In ΔPQR, ray QS bisects $\angle Q$ with S lying on side PR. If $PQ = 14$, $QR = 21$, and $PR = 25$, find the length of segment PS.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Trigonometry

Find the equation of the line that passes through $(4,\ -1)$ and is perpendicular to the line $2x - 3y = 9$. Express your answer in slope-intercept form.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Transformations & Vectors

A projectile is launched from ground level with height modelled by $h(t) = -16t^2 + 64t + 80$ feet after t seconds. Find (a) the maximum height reached, and (b) the time when it hits the ground.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Functions

In a geometric sequence, the 3rd term is 18 and the 6th term is 486. Find the common ratio r, the first term a_1, and the sum of the first 5 terms S_5.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number

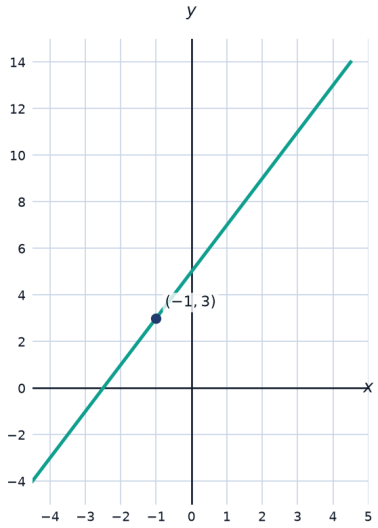
Solve the nonlinear system $y = x^2 - 3x + 2$ and $y = 2x - 4$ algebraically. State all intersection points (x, y).

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in multiplying and dividing with indices, evaluate $f(2)$.



a. 9

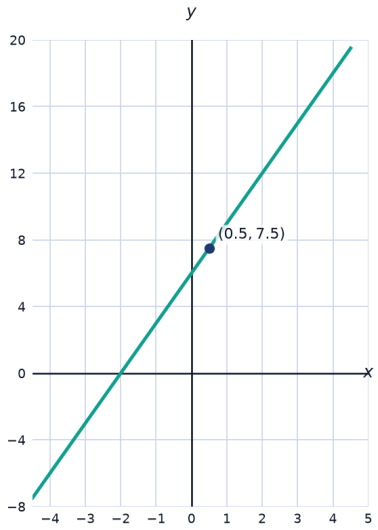
b. 11

c. 4

d. 13

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in index laws, evaluate $f(2)$.



a. 12

b. 15

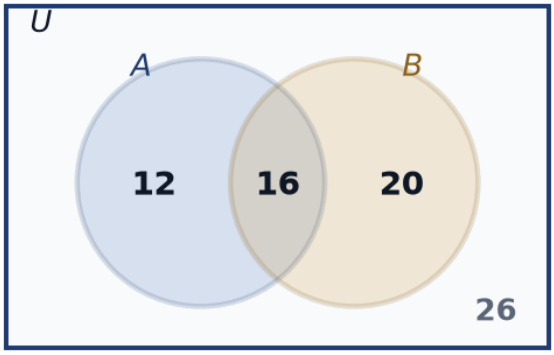
c. 6

d. 16

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for multiplying and dividing with indices consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?



a. 16.5

b. 17.5

c. 18.5

d. 20.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for index laws consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a grouped frequency table, a class interval is $20 \le x < 35$ with a frequency of 30. What is the frequency density of this bar on a histogram?

a. 1.5

b. 2.0

c. 0.5

d. 15

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn at random without replacement. What is the probability that BOTH marbles are red?

a. $\frac{20}{56}$

b. $\frac{25}{64}$

c. $\frac{5}{8}$

d. $\frac{22}{56}$

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two unbiased 6-sided dice are rolled simultaneously. What is the probability that the sum of the numbers shown is equal to 9?

a. $\frac{1}{9}$

b. $\frac{1}{6}$

c. $\frac{5}{36}$

d. $\frac{1}{12}$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a test, the mean mark of 15 girls was 72, and the mean mark of 10 boys was 68. What was the combined mean mark of all 25 students?

a. 70.4

b. 70.0

c. 70.8

d. 69.6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the interquartile range (IQR) of the following set of data: 4, 7, 9, 12, 15, 18, 22.

a. 11

b. 7

c. 18

d. 14

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two independent events A and B have $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{2}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{7}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{3}{4}$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The probability that it rains on any given day is 0.3. What is the probability that it rains on at least one of the next two days?

a. 0.51

b. 0.60

c. 0.09

d. 0.49

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The numbers 3, 5, 7, 8, x, 12 have a mean of 8. What is the value of x?

a. 13

b. 11

c. 14

d. 9

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student draws a pie chart to represent favourite sports. If 18 out of 72 students chose Basketball, what is the sector angle representing Basketball?

a. 90°

b. 45°

c. 72°

d. 120°

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cumulative frequency diagram is plotted for 80 students. At what cumulative frequency value do you read the median score?

a. 20

b. 40

c. 60

d. 80

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.7$, find $P(A \cap B)$.

a. 0.28

b. 1.10

c. 0.30

d. 0.55

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Events A and B are mutually exclusive. If $P(A) = 0.35$ and $P(B) = 0.45$, find $P(A \cup B)$.

a. 0.80

b. 0.1575

c. 0.70

d. 0.10

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A box contains 4 red balls, 5 green balls, and 3 yellow balls. One ball is chosen at random. What is the probability that it is NOT green?

a. $\frac{7}{12}$

b. $\frac{5}{12}$

c. $\frac{1}{3}$

d. $\frac{3}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Transformations & Vectors

On a cumulative frequency curve representing marks of 120 students, the median mark corresponds to which cumulative frequency?

a. 60

b. 30

c. 90

d. 100

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Functions

In a survey of 20 people, 10 own a car. Of those, 5 also own a bicycle. Find the probability that a randomly chosen car owner also owns a bicycle.

a. $\frac{3}{7}$

b. $\frac{1}{7}$

c. $\frac{2}{7}$

d. $\frac{1}{2}$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number

A spinner has probability 0.35 of landing on Gold. If the spinner is spun 200 times, what is the expected number of times it lands on Gold?

a. 70

b. 80

c. 100

d. 65

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A bag has 3 red and 2 blue balls. Two balls are drawn without replacement. Find the probability both are red.

a. $\frac{1}{10}$

b. $\frac{3}{10}$

c. $\frac{1}{2}$

d. $\frac{4}{5}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Algebra & Graphs

The probability that it rains tomorrow is $\frac{4}{5}$. Find the probability that it does NOT rain.

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $\frac{1}{5}$

d. $\frac{2}{5}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Coordinate Geometry

From a cumulative frequency graph, the lower quartile Q_1 is 28 and the upper quartile Q_3 is 54. Calculate the interquartile range (IQR).

a. 26

b. 28

c. 41

d. 82

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

The frequency table shows ages of 50 museum visitors. Calculate (a) mean age, (b) modal class.

a. Mean ≈ 39, Modal class 40–59

b. Mean ≈ 42, Modal class 40–59

c. Mean ≈ 39, Modal class 20–39

d. Mean ≈ 42, Modal class 20–39

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry

A data set has values and frequencies: $1(\times 4), 2(\times 3), 3(\times 2), 4(\times 1)$. Find the mean.

a. 5

b. 2

c. 11

d. 10

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